

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Dr.V.Parthipan

Code of Conduct:

I NS BHAVESH (192565086) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student

NS BHAVESH

Criteria	Understanding of Concepts 2.5	Experimental Design 3	Use of Tools 2	Analysis of Results 1.5	Documentation 1	Total (10)
Q1	2.5	3	2	1.5	1	10
Q2	2.5	3	2	1	1	9.5
Q3						
Q4						

Faculty Signature

Experiments

QUESTIONS:4

Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP.(BL4) (10)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Q:1

The image shows a Wireshark packet capture of a TCP connection. The filter is set to `tcp.flags.syn == 1 || (tcp.flags.ack == 1 && tcp.len == 0)`. The packet list shows several packets, with packet 194 selected. The packet details pane shows the following information:

- Frame 194: Packet, 66 bytes on wire (528 bits), 66 bytes captured (528 bits) on interface \Device\NPF_{5C347B90-9CE3-4B84-BAG4-354589838239}, id 0
- Ethernet II, Src: Netgear_Gc:73:e2 (80:cc:9c:6c:73:e2), Dst: AzureWaveTec_72:91:20 (ec:3a:56:72:91:20)
- Internet Protocol Version 4, Src: 3.254.238.10, Dst: 172.23.18.185
- Transmission Control Protocol, Src Port: 443, Dst Port: 61066, Seq: 0, Ack: 1, Len: 0
- Source Port: 443
- Destination Port: 61066
- [Stream index: 12]
- [Stream Packet Number: 2]
- [Conversation completeness: Incomplete, DATA]
- [TCP Segment Len: 0]
- Sequence Number: 0 (relative sequence number)
- Sequence Number (raw): 2017844721
- [Next Sequence Number: 1 (relative sequence number)]
- Acknowledgment Number: 1 (relative ack number)
- Acknowledgment number (raw): 4111249062
- Flags: 0x012 (SYN, ACK)
- Window: 65340
- [Calculated window size: 65340]
- Checksum: 0xd4c2 [unverified]
- [Checksum Status: Unverified]
- Urgent Pointer: 0
- Options: (12 bytes), Maximum segment size, [Timestamps]
- [SEQ/ACK analysis]
- [Client Contiguous Streams: 1]
- [Server Contiguous Streams: 1]

The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, and Transmission Control Protocol header.

Q:2

The image shows a Wireshark packet capture of a DNS query and response. The filter is set to `dns`. The packet list shows several packets, with packet 1222 selected. The packet details pane shows the following information:

- Frame 1222: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits) on interface \Device\NPF_{5C347B90-9CE3-4B84-BAG4-354589838239}, id 0
- Ethernet II, Src: AzureWaveTec_72:91:20 (ec:3a:56:72:91:20), Dst: Cisco_6d:bc:f7 (24:d5:e4:6d:bc:f7)
- Internet Protocol Version 4, Src: 172.23.18.185, Dst: 172.21.100.31
- User Datagram Protocol, Src Port: 58692, Dst Port: 53
- Source Port: 58692
- Destination Port: 53
- Length: 36
- Checksum: 0x3750 [unverified]
- [Checksum Status: Unverified]
- [Stream index: 210]
- [Stream Packet Number: 1]
- [Timestamps]
- UDP payload (28 bytes)
- Domain Name System (query)
- Transaction ID: 0x0003
- Flags: 0x0100 Standard query
- Questions: 1
- Answer RRs: 0
- Authority RRs: 0
- Additional RRs: 0
- Queries
- [Response In: 1278]

The packet bytes pane shows the raw data of the packet, including the Ethernet II header, Internet Protocol Version 4 header, User Datagram Protocol header, and Domain Name System (query) header.

Q:3

The image shows a Wireshark packet capture window titled "Wi-Fi". The packet list pane on the left shows a series of TCP retransmissions. The packet details pane on the right shows the structure of a selected packet (No. 183), which is a TCP segment. The packet structure is as follows:

- Frame 183: Packet, 78 bytes on wire (624 bits), 78 bytes captured
- Ethernet II, Src: Netgear_Gc73:e2 (80:cc:9c:6c:73:e2), Dst: Azurel
- Internet Protocol Version 4, Src: 3.253.179.213, Dst: 172.23.18.185
- Transmission Control Protocol, Src Port: 443, Dst Port: 52070, Seq: 4294967266, Win: 187, Len: 31
- Transport Layer Security

The packet list pane shows the following details:

No.	Time	Source	Destination	Protocol	Length	Info
183	4.096169100	3.253.179.213	172.23.18.185	TLSv1.2	78	[TCP Retransmission], Application Data
225	4.669223100	15.217.179.25	172.23.18.185	TLSv1.2	100	[TCP Retransmission], Application Data
226	4.669224000	15.217.179.25	172.23.18.185	TCP	85	[TCP Retransmission] 443 → 65467 [PSH, ACK] Seq=4294967266 Ack=1 Win=187 Len=31
231	4.674089300	51.92.249.166	172.23.18.185	TLSv1.2	78	[TCP Retransmission], Application Data
235	4.677358700	51.92.249.166	172.23.18.185	TLSv1.2	78	[TCP Retransmission], Application Data
283	5.941212600	35.154.1.3	172.23.18.185	TLSv1.2	93	[TCP Retransmission], Application Data
284	5.941213000	35.154.1.3	172.23.18.185	TCP	78	[TCP Retransmission] 443 → 65034 [PSH, ACK] Seq=4294967273 Ack=1 Win=198 Len=24
340	7.475189000	108.130.186.25	172.23.18.185	TLSv1.2	78	[TCP Retransmission], Application Data
1255	27.438078800	150.171.110.85	172.23.18.185	TLSv1.2	93	[TCP Retransmission], Application Data
1256	27.438072000	150.171.110.85	172.23.18.185	TCP	78	[TCP Retransmission] 443 → 64804 [PSH, ACK] Seq=40 Ack=2 Win=191 Len=24
1600	36.042142200	150.171.110.82	172.23.18.185	TCP	54	[TCP Retransmission] 443 → 50763 [FIN, ACK] Seq=64 Ack=1 Win=147 Len=0
2494	57.959378400	23.199.67.57	172.23.18.185	TLSv1.2	78	[TCP Retransmission], Application Data
2495	57.959379400	23.199.67.57	172.23.18.185	TCP	54	[TCP Retransmission] 443 → 52075 [FIN, ACK] Seq=25 Ack=2 Win=602 Len=0
2549	58.987285700	23.199.67.57	172.23.18.185	TLSv1.2	78	[TCP Retransmission], Application Data
2550	58.987286600	23.199.67.57	172.23.18.185	TCP	54	[TCP Retransmission] 443 → 55378 [FIN, ACK] Seq=25 Ack=2 Win=492 Len=0
2554	59.209148300	172.23.18.185	23.199.67.57	TCP	54	[TCP Retransmission] 55378 → 443 [FIN, ACK] Seq=2 Ack=26 Win=255 Len=0

Q:4

The image shows a Wireshark packet capture window titled "Capturing from Wi-Fi". The packet list pane on the left shows a series of DNS packets. The packet details pane on the right shows the structure of a selected packet (No. 44), which is a DNS query response. The packet structure is as follows:

- Frame 44: Packet, 98 bytes on wire (784 bits), 98 bytes captured
- Ethernet II, Src: Cisco_Ed6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: AzureWaveTec_72:91:20 (ec:3a:56:72:91:20)
- Internet Protocol Version 4, Src: 172.21.100.31, Dst: 172.23.18.185
- User Datagram Protocol, Src Port: 53, Dst Port: 51242
- Domain Name System (response)

The packet list pane shows the following details:

No.	Time	Source	Destination	Protocol	Length	Info
41	2.842278800	172.23.18.185	172.21.100.31	DNS	86	Standard query PTR 31.100.21.172.in-addr.arpa
42	2.865229700	172.21.100.31	172.23.18.185	DNS	115	Standard query response 0x0001 PTR 31.100.21.172.in-addr.arpa PTR s3.saveetha.net
43	2.870424000	172.23.18.185	172.21.100.31	DNS	86	Standard query PTR 31.100.21.172.in-addr.arpa
44	2.876648100	172.21.100.31	172.23.18.185	DNS	115	Standard query response 0x0001 PTR 31.100.21.172.in-addr.arpa PTR s3.saveetha.net
764	22.425767700	192.168.0.239	172.23.18.185	DNS	86	Standard query PTR 31.100.21.172.in-addr.arpa

The packet details pane shows the structure of the selected packet (No. 44):

- Transaction ID: 0x0002
- Flags: 0x1800 Standard query response, No error
- Questions: 1
- Answer RRs: 1
- Authority RRs: 0
- Additional RRs: 0
- Queries
- Answers

The packet bytes pane shows the raw data of the selected packet (No. 44):

```
0000  ec 3a 56 72 91 20 00 00 00 00 00 00 00 00 00 00  ..Vr.....E-
0010  00 54 22 47 00 00 00 00 00 00 00 00 00 00 00 00  -T?.....d-
0020  12 b9 00 35 c8 2a 00 40 1e 02 00 02 81 80 00 01  -..5.*.@.....
0030  00 01 00 00 00 00 06 67 6f 6f 67 6c 65 03 63 6f  -.....oogle.co
0040  6d 00 00 1c 00 01 c0 0c 00 1c 00 01 00 00 01 29  -m.....
0050  00 10 24 04 68 00 40 07 00 33 00 00 00 00 00 00  -..$.h@.....
0060  20 0e
```